

# HCWC Distribution Builder

From geological mechanisms and DHIs to contact distribution, risk and much more — all in one model.

by Lars Hjelm

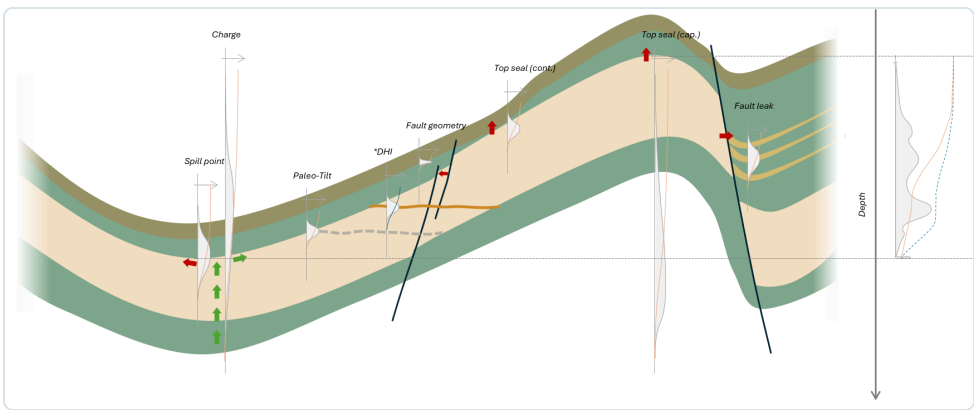
Build the HCWC distribution from competing geological limits.  
Read the probability of success at every depth off the same sample.  
Update with DHI evidence without discarding the geology.



## 1 The idea

Don't specify the HCWC — derive it from the geology.

- The tool models the **mechanisms that can stop a column**. Each defines its own limit.
- In every realisation all limits are sampled; the **shallowest active** one sets the contact.
- Across many realisations the distribution emerges from the competition — not from a shape somebody chose.
- A DHI enters afterwards as **evidence over the same realisations** (panels 4 and 5), never as a replacement contact.



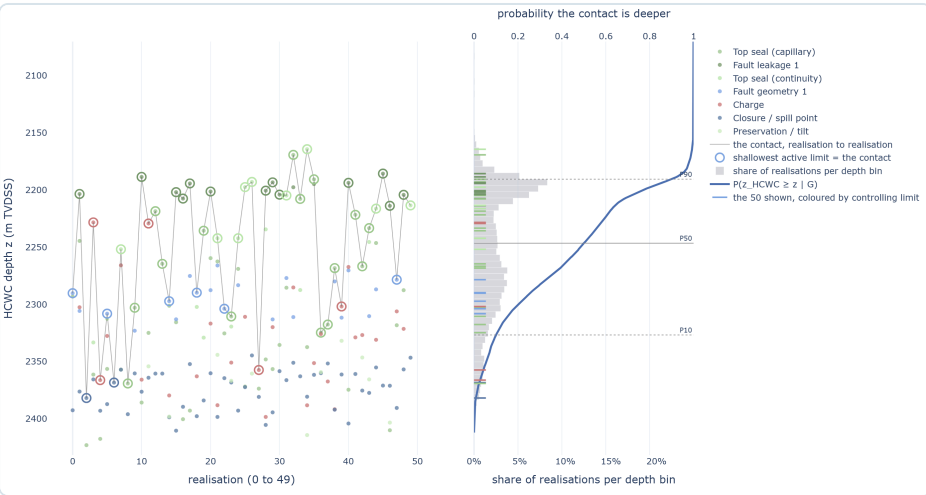
Every mechanism on one section, with the depth distribution at which it acts. Capacities are measured from the apex. Figure by Lars Hjelm.

LIMITS, COLOURED BY THE RISK ELEMENT THEY BELONG TO

- Structural spill** · closure — what the trap geometry permits
- Fault geometry** · closure — juxtaposition across the bounding fault
- Charge limitation** · charge — not enough charge to fill deeper
- Capillary seal capacity** · retention — the column held before breakthrough
- Seal continuity** · retention — a channel, window or thinning
- Fault leakage** · retention — gouge, fault-rock, reactivation
- Mechanical failure** · retention — stress headroom, not entry pressure
- Reservoir pinch-out** · reservoir — connected pore volume ends

## 2 How the tool works

- Define the prospect**  
Apex, spill, the element chances giving  $P(G)$ , the assessment minimum, the well. Tab 2.0.
- Elicit each limit**  
A capacity in metres below the crest, a mapped surface as a depth — whichever the geologist knows. Tab 3.0.
- Run the competition**  
Each limit drawn for presence and depth; the shallowest active one sets the contact **and is recorded**. Tab 4.0.
- Add DHI evidence — optional**  
The same realisations are reweighted. No second model, no arbitrary "DHI case". Tab 5.0.



The competition, realisation by realisation: every limit's sampled depth, the shallowest ringed in its own colour, and the distribution those minima make.

$$H = \min\{h_i : \text{limit } i \text{ active}\}$$

Minimum, not a blend — a limit only changes realisations that reach it. Blending a deep leak into a background distribution can make a prospect look *larger*.

## 4 Character vs geometry

One observation, two kinds of evidence — each updating exactly one factor.

The two answer different questions and have different consequences. A bright, conformable response raises the chance that *anything* is there without saying where the column ends; a confidently picked flat spot says where it ends without saying whether it is hydrocarbons. Collapsed into one number, a single judgement has to do both jobs — and the usual result is a contact moved on the strength of amplitude character, or a chance raised on the strength of a pick.

### CHANNEL 1 · CHARACTER

#### Are hydrocarbons present?

Amplitude, polarity, conformity, offset behaviour, placed on the evidence index. Updates the chance; **never moves the contact**.

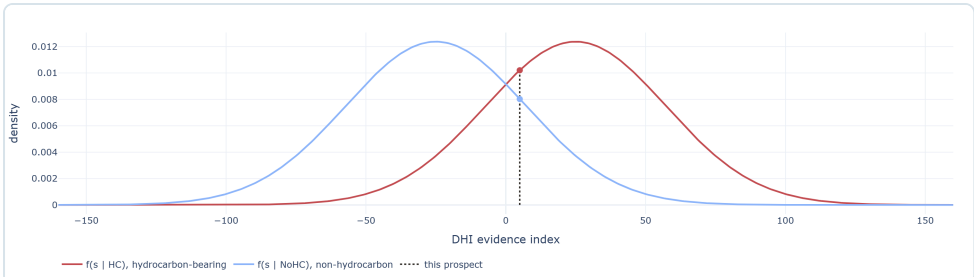
$$LR(s) = \frac{f(s|HC)}{f(s|NoHC)}$$

### CHANNEL 2 · GEOMETRY

#### Where does the column end?

The picked depth and the confidence in it, as a likelihood over realisations already drawn. Reshapes the contact; **never moves  $P(G)$** .

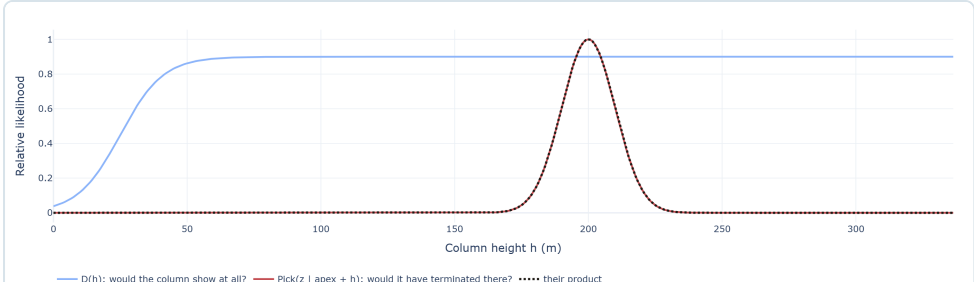
$$L = c \cdot D(h) \cdot \text{Pick}(z) + (1-c) \cdot s$$



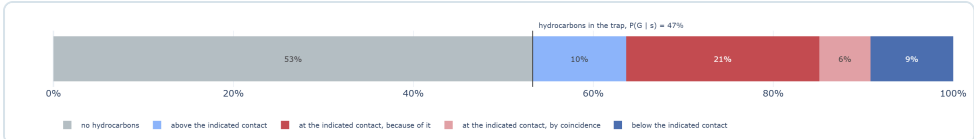
The two reference distributions on the index. The likelihood ratio is the ratio of the two marked heights — which is why the axis units do not matter.



The ratio against the index, the posterior it gives, and the same update across priors — neutral evidence returning the prior on the diagonal.



$D(h)$  asks whether a column that size would have shown at all; the pick likelihood asks whether it would have terminated where the observed one did. Their product is the geometry evidence.

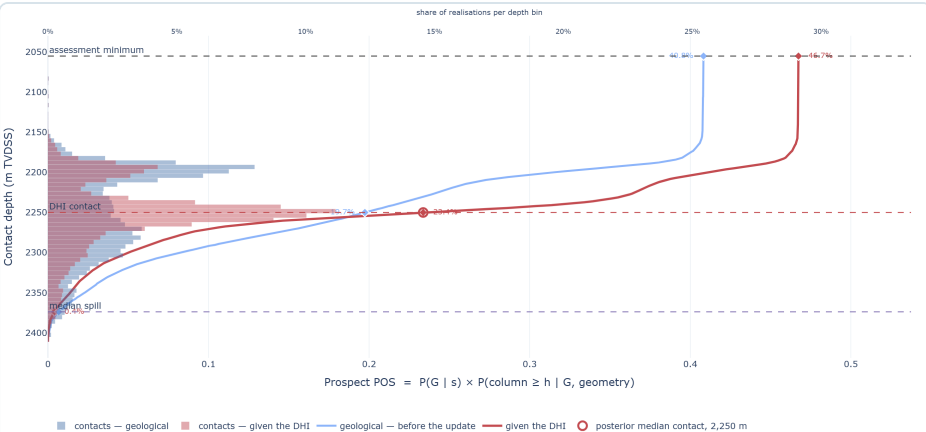


What a seen indication can turn out to have been, in depth order, as shares of *all* outcomes: no hydrocarbons at all; or — sharing the updated chance — a contact above the indicated band, within it because the indication *is* the contact, within it by coincidence, and below it.

## 5 DHI — update, don't replace

A seismic indication is evidence, not a separate case.

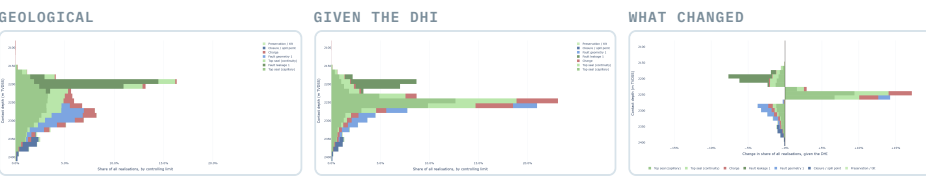
- The existing geological realisations are **reweighted**, not re-simulated.
- Each keeps its controlling limit, so prior and posterior are comparable figure for figure.
- How far the answer moves follows from **the strength of evidence stated**, not from a decision to believe the DHI.



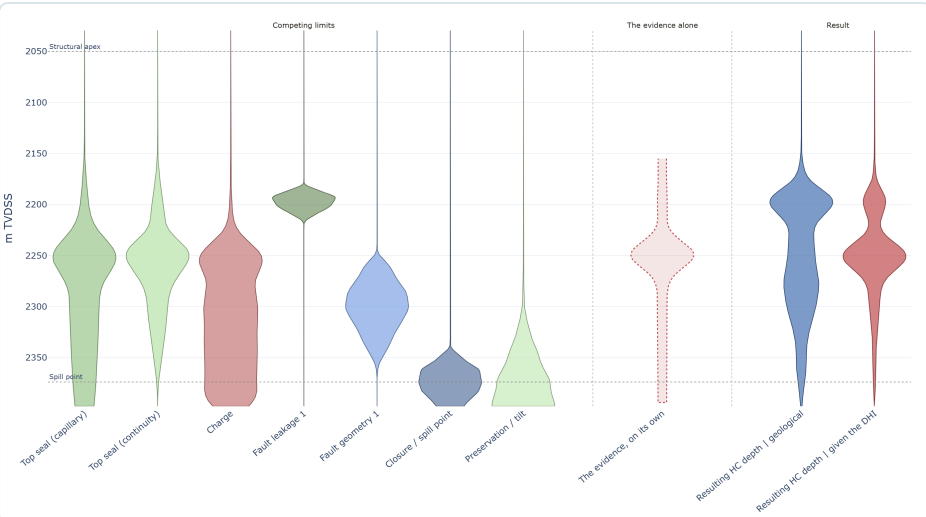
The chance against threshold, geological and updated. The index *scales* the curve; the geometry *reshapes* it. The open circle is the posterior median, landing on the pick.

$$w_j \propto L(D | h_j, G)$$

Self-normalised importance sampling: the posterior is the same sample, reweighted.



The controlling mechanism at each depth, read three ways. Bin height carries the contact distribution, so the third panel — gains right of the line, losses left — shows the evidence pulling realisations into the indicated band and taking them from above and below it. **The evidence moves which depths are reached far more than it moves the mechanism mix at any one depth**, and the element chances on tab 2.0 are untouched.

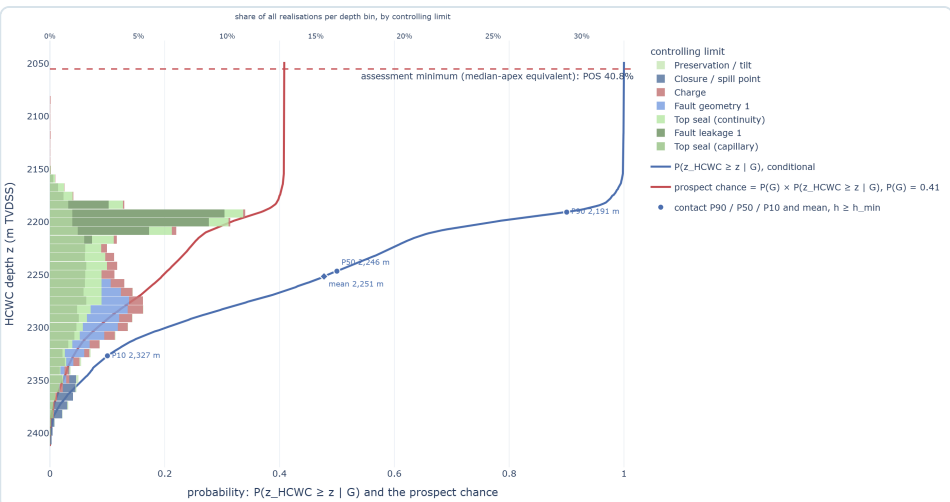


Every limit as a violin on one depth axis. The evidence sits on its own in the middle, drawn hollow because it is a likelihood rather than a count of realisations — its shape carries the information, not its area. Both answers are on the right.

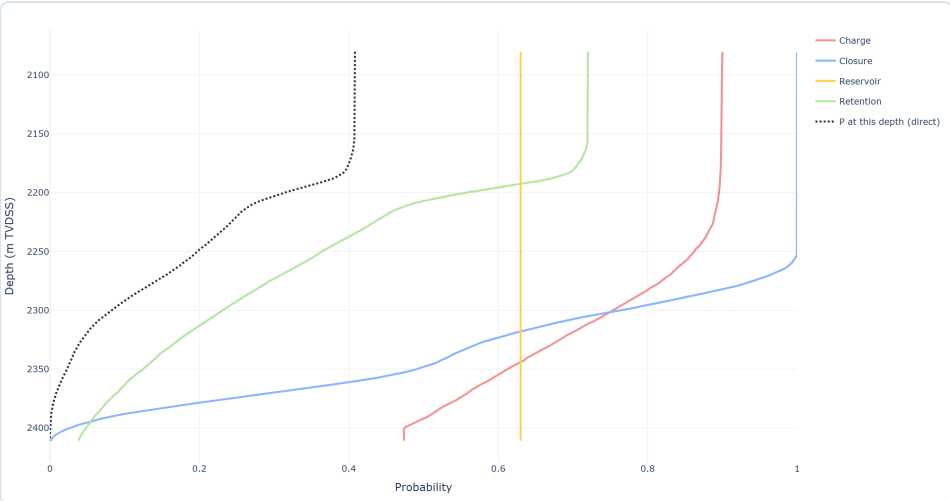
**What it is not.** Substituting a DHI contact depth treats an observation as a scenario — and a scenario cannot be argued with on geological grounds.

## 3 Key outputs

- The **HCWC distribution**, and which mechanism controlled each realisation.
- Probability of success at every depth**, read at the assessment minimum and at the well.
- A **curve per risk element**, for the volumetric tool downstream.
- All of it off **one sample**, so there is nothing to reconcile afterwards.



Blue is  $P(z_{HCWC} \geq z | G)$ ; red is the prospect chance,  $P(G)$  times it — the gap is the element risk. Bars are the controlling limit per depth bin.



Each element's curve, derived from its own shallowest active limit rather than allocated by judgement.

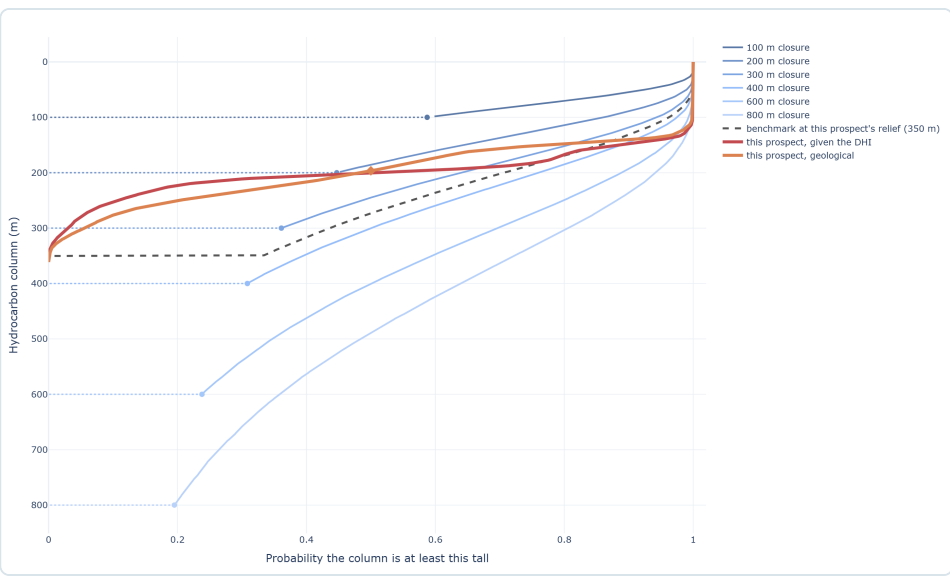
## 6 The empirical check

Filled-to-spill discoveries are right-censored, not measurements.

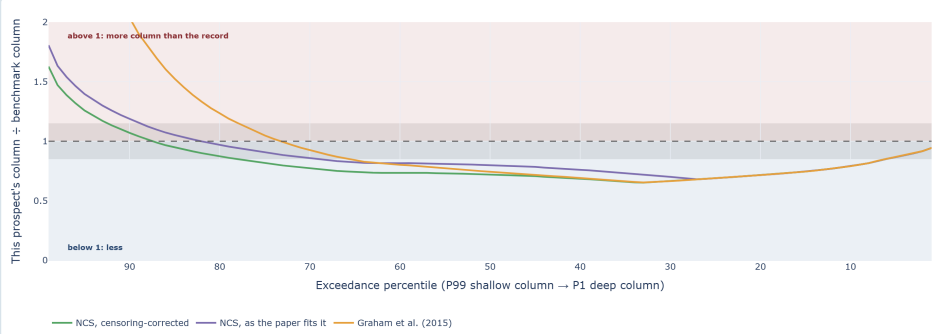
- Such a pool says the seal held **at least** the trap height — not what it could have held.
- Treated as exact values they bias a fit toward structural spill, **dropping them trades one bias for another**.
- The record is **compared, never joined**. A disagreement is a finding to explain, not a correction to apply.



Red diamonds lie on the 1:1 line by definition — they record the closure, not the seal. The censored fit sits below the ordinary one because it estimates seal capacity.



Orange is the prospect; the dashed grey beside it the benchmark at the same relief. Those two are the pair — the coloured curves are the family it sits inside, not its comparators.



The slope is the **finding**, not the **level**. A flat curve off parity is one correctable bias; a tilt is a disagreement about shape, which no single correction fixes.

## 7 Keep these three apart

Merging them is where most of the confusion in this area comes from. Every probability the app prints says which it is, whether it includes  $P(G)$ , and the threshold it was read at.

| QUANTITY                    | WHAT IT IS                                                                  | CONDITIONAL ON  |
|-----------------------------|-----------------------------------------------------------------------------|-----------------|
| $P(G)$                      | An accumulation exists at all — the elements worked at the crest.           | Nothing         |
| $F(h) = P(H \geq h   G)$    | The column reaches $h$ , given an accumulation. What the limits produce.    | An accumulation |
| $POS(h) = P(G) \times F(h)$ | Hydrocarbons found at that depth — the product, and what an explorer wants. | Nothing         |

**The commonest error.** Quoting the conditional curve as the prospect's chance sets  $P(G)$  to 1 and overstates the prospect by  $1 / P(G)$  — largest exactly where the geology is least certain.

### Monotonic by construction

A deeper contact cannot be likelier than a shallower one — every point is read off one sample. A band-by-band array has no such guarantee.

### Uncertainty is not influence

A wide range can rarely decide the answer; a narrow one that usually decides it moves everything. Both are reported, separately.

### Know what the answer rests on

Effective sample size says how concentrated the weighting became; the sensitivity says which input carries it. **Where a typed DHI number moves the answer further than the geology does, the posterior is a statement about the seismic assumptions rather than about the prospect.**

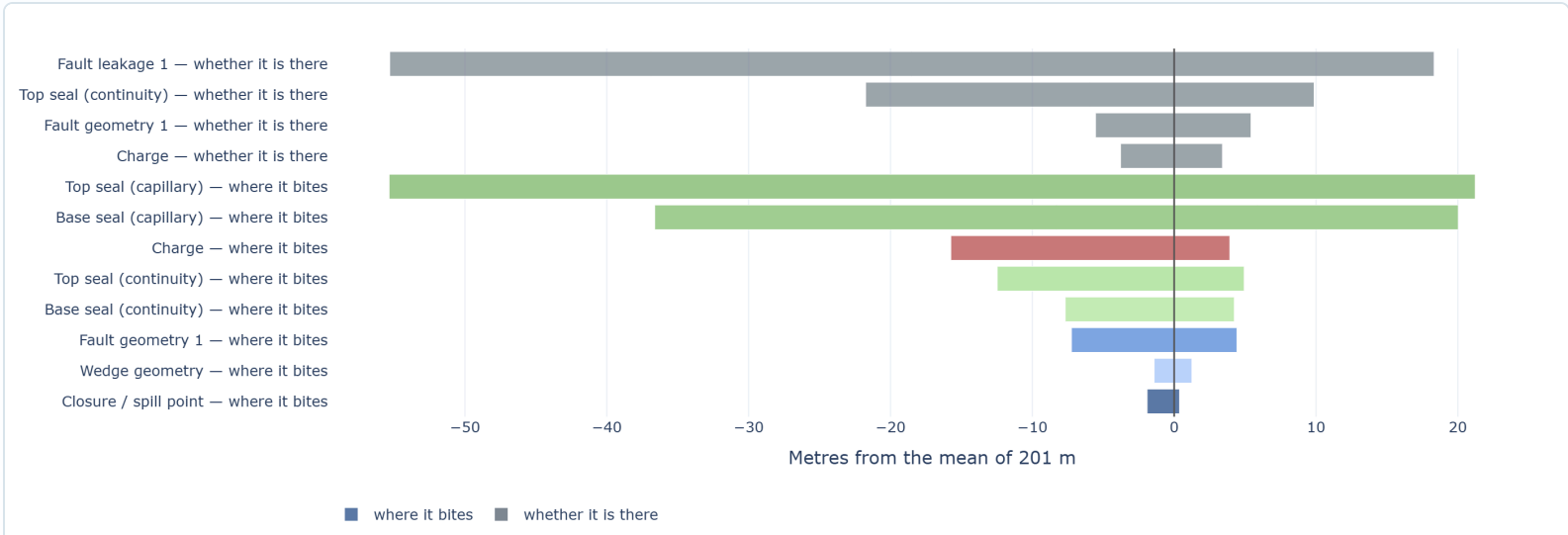
### Absence is evidence, conditionally

A missing anomaly counts only in so far as that column would have been detectable. The ceiling on  $D(h)$  is below 1 by construction.

## 8 Where the elicitation effort belongs

How often a limit controls the contact is not how much it moves the answer.

- A limit can set the contact in most realisations and still barely move the mean, if its range is narrow.
- A limit that rarely controls can move it a long way, if the range it would impose is wide.
- Elicitation effort belongs on the bars at the top; a limit near zero can stay at a rough value.



How much each elicited number moves the mean, on the geological model. Two kinds of bar: *where* a limit applies is its distribution, *whether it* is there is its  $P(\text{active})$ . Each bar is the mean with input in its top tenth against its bottom tenth.

**Read it before the elicitation, not after.** It says which two or three numbers are worth a workshop, and which can stay as they are.

## 9 The whole model, in one line

### Risk elements

Spill, charge, seal capacity and continuity, fault leakage, mechanical failure, pinch-out.

$$P(G) = \prod P_e$$

### Competing limits

Spill, charge, seal capacity and continuity, fault leakage, mechanical failure, pinch-out.

$$H = \min\{h_i\}$$

### HCWC distribution

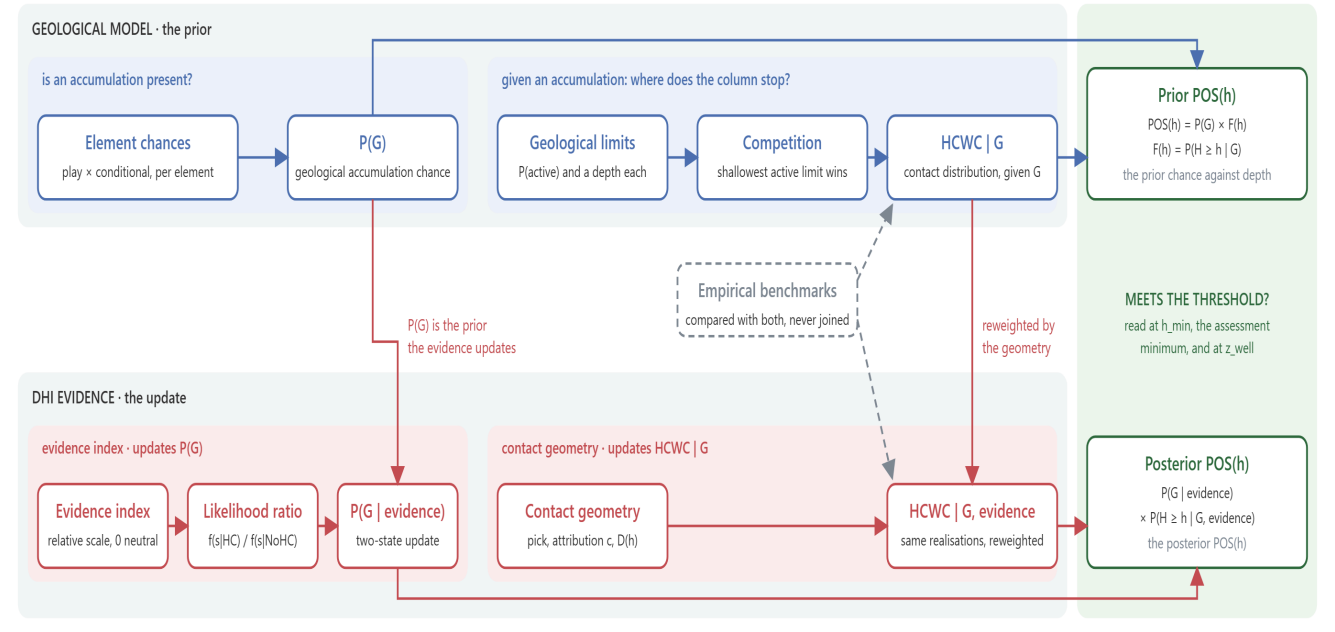
Emerges from the competition, with the controlling mechanism recorded.

$$F(h) = P(H \geq h | G)$$

### Probability of success

The product, read at the minimum and at the well — then updated by evidence.

$$POS(h) = P(G) \cdot F(h)$$



The geological model is the prior; the DHI is evidence. The index never moves the contact, the geometry never moves  $P(G)$ , each enters once, and one weighted sample reads every posterior quantity.

**What it is not.** Not a basin, hydrodynamic or 3D flow model. Where a more detailed model exists, its output enters as *one more competing limit*.

### WHERE TO GO NEXT

**Run the tool**  
hcwc-builder.streamlit.app

**Read the source**  
github.com/lhjelm-dk/HCWC-Builder

**The derivations**  
tab 8.2 of the tool

### NOTATION

$G$  the accumulation state ·  $H$  column height, m ·  $z_{\text{apex}}$  apex depth, m TVDSS ·  $z_{\text{HCWC}}$  =  $z_{\text{apex}} + H$  contact depth, increasing downward ·  $h_{\text{min}}$  the assessment minimum, a column height ·  $s$  the DHI evidence index ·  $G$  the contact attribution ·  $D(h)$  detectability ·  $LR$  likelihood ratio ·  $w_j$  the weight on realisation  $j$ . A probability written  $| G$  is conditional on the accumulation.

### REFERENCES · THE FULL BIBLIOGRAPHY IS TAB 8.1.11

Beha, A., Christensen, J. E. & Young, R. (2012) — A general method for the consistent volume assessment of complex hydrocarbon traps. Journal of Petroleum Geology **35**(1), 85–97. doi:10.1111/j.1747-5457.2012.00520.x

Hood, K. C. (2024) — Hydrocarbon Column Heights, Part 1 and Part 2. Rose & Associates blog, 7 May 2024, from Hood (2019), Risk Coordinators Workshop #17, Houston.

Edmundson, I., Davies, R., Frette, L. U., Mackie, S., Kavli, E. A., Rotevatn, A., Yielding, G. & Dunbar, A. (2021) — An empirical approach to estimating hydrocarbon column heights for improved predrill volume prediction in hydrocarbon exploration. AAPG Bulletin **105**(12), 2381–2403. doi:10.1306/03122119223

Monigle, P. W., Hedayati, T. S. & Goulding, F. J. (2025) — Integrated and improved direct hydrocarbon indicators: a step forward in petroleum risk discrimination. AAPG Bulletin **109**(5), 617–636. doi:10.1306/04042524030

Lowry, D. C., Suttill, R. J. & Taylor, R. J. (2005) — Advances in risking exploration prospects. The APPEA Journal **45**(1), 143–158. doi:10.1071/AJ040412

Not in that literature, and added here: the per-element depth-dependent chance, the likelihood form of DHI evidence, and the censoring correction to the empirical record. Every figure on this sheet is the app's own output, exported from the worked prospect that ships with the tool.